CardioRisk: an end-to-end MLOps pipeline for heart-disease risk prediction

AIMLCZG523 Machine Learning Operations, Assignment 1

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Code repository: https://github.com/shubham-pardeshi26/Bits-Sem3-MLOps-Assignment1/

Charts and the architecture diagram are generated by the code (`make eda`, `make train`, `scripts/architecture_diagram.py`). Of the screenshots (`screenshots/01` to `15`), the five that prove CI/CD and deployment are in the body and the rest in Appendix C.

1. Summary

Problem. Predict whether a patient has heart disease (UCI attribute `num > 0`) from 13 routine clinical measurements, and serve the model as a monitored API that runs unchanged on a laptop, in CI and on Kubernetes.

Approach. Most work on this dataset uses only the 303 Cleveland patients. I used all four UCI hospitals (Cleveland, Hungary, Switzerland, VA Long Beach): three times the data, but each site recorded different tests and the disease rate ranges from 36% to 93%. Missing data and hospital shift become the central problems, and the data allows a test at a hospital the model has never seen. The code is one Python package, `cardiorisk`, with one `make` target per stage; CI runs the same commands.

Table 1. Headline results (from `reports/metrics.json` and `models/model_card.json`).

Item	Result
Data	918 patients from 4 hospitals (920 raw rows, 2 duplicates dropped), 55% with heart disease
Model search	Majority baseline, logistic regression, random forest, histogram gradient boosting: 85 grid candidates, 5-fold CV
Champion	Elastic-net logistic regression (`C = 0.3`, `l1_ratio = 0.5`): CV ROC-AUC 0.907 ± 0.029 on 734 training patients
Hold-out (184 patients, threshold 0.5)	ROC-AUC 0.906, accuracy 0.842, precision 0.829, recall 0.902, F1 0.864
Unseen hospital (leave-one-site-out ROC-AUC)	Cleveland 0.876, Hungary 0.902, Switzerland 0.749, VA 0.730
MLOps	MLflow tracking and registry (`cardiorisk-classifier@champion`); FastAPI in a non-root image; Kubernetes with Kustomize overlays, LoadBalancer, Ingress and HPA 2-5; Prometheus and Grafana; GitHub Actions CI/CD publishing to GHCR

2. Architecture

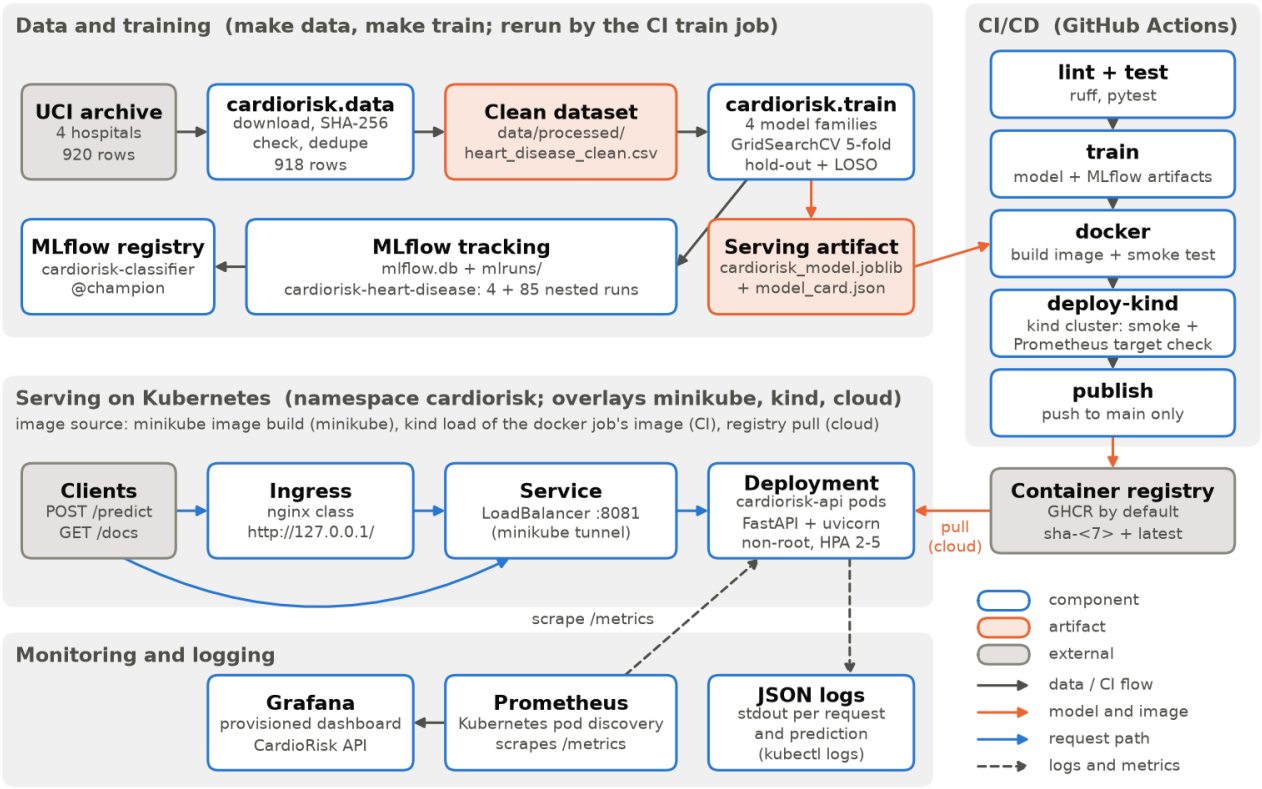


Figure 1. System architecture: data and training, CI/CD, serving on Kubernetes, and monitoring.

Figure 1 shows the four parts: data and training (checksummed data, grid search tracked in MLflow, a serving artifact), CI/CD (retrain, test in Docker and kind, publish from `main`), serving (FastAPI pods behind a LoadBalancer and an Ingress) and monitoring (Prometheus, Grafana, JSON logs). Each concern has one code path: the cross-validated pipeline is the served object, the API and the CLI share `predict_records`, and the tested image is the published one.

3. Setup and installation

Prerequisites: Python 3.12 and `make`; Docker for the container; minikube and kubectl for Kubernetes; Google Chrome only for the PDF of this report.

```
make setup          # .venv with the pinned requirements.txt
make data           # verify (or download) the UCI files, write the cleaned CSV
make eda            # EDA figures in reports/figures/eda_*.png
make lint test      # ruff and the 39 unit tests, as in CI
make train          # tune, track in MLflow, save the champion (about 30 s)
make mlflow-ui      # MLflow UI on http://127.0.0.1:5001
make serve          # API on http://127.0.0.1:8081/docs
make docker-build docker-run   # then, in a second terminal: make smoke
make compose-up     # API + Prometheus :9090 + Grafana :3000 without Kubernetes
make minikube-start k8s-image k8s-deploy
make report report-pdf         # this report: reports/report.docx and .pdf
```

The API uses host port 8081 and MLflow 5001, because 8000, 8080 and 5000 are often taken; `PORT=` and `MLFLOW_PORT=` change them. Every Kubernetes target runs `kubectl --context minikube`, and `make k8s-deploy` stops if that context is missing. The README has the full command reference and troubleshooting.

4. Data acquisition and EDA

4.1 Source and cleaning

The UCI Heart Disease dataset (id 45) ships one "processed" file per hospital with the same 14 attributes: Cleveland (303 rows), Hungary (294), Switzerland (123) and VA (200). `make data` checks the files against SHA-256 digests pinned in `data.py` and downloads the archive only if one is missing or differs, since a silently republished archive would shift every number here. Raw files and the cleaned CSV are committed, so CI never depends on the UCI server. `clean()` applies the rules in Table 2, each unit-tested. A `site` column is kept for stratification and leave-one-site-out evaluation, never as an input.

Table 2. Cleaning rules and how many values they touch.

Rule	Affected	Reason
? read as missing	1,759 values	UCI marker for "not recorded"
`chol == 0` set to missing	172 rows (all 123 Swiss, 49 VA)	Zero cholesterol is impossible: "not measured"
`trestbps == 0` set to missing	1 row (VA)	Same reason
Exact duplicates dropped	2 rows (1 Hungary, 1 VA)	A copy could land on both sides of the split
`target = (num > 0)`	all 918 rows	Hungary coded `num` only as 0/1, the others graded 1-4
Kept: negative `oldpeak` (12 rows), extremes such as `chol = 603`	-	Clinically real, so no clipping

Imputation is not cleaning: it is part of the model pipeline, fitted on training folds only.

4.2 Exploratory analysis

`make eda` draws the figures and `notebooks/01_eda.ipynb` discusses them; Figure 2 is here, the others are in Appendix B. Each finding led to a pipeline decision.

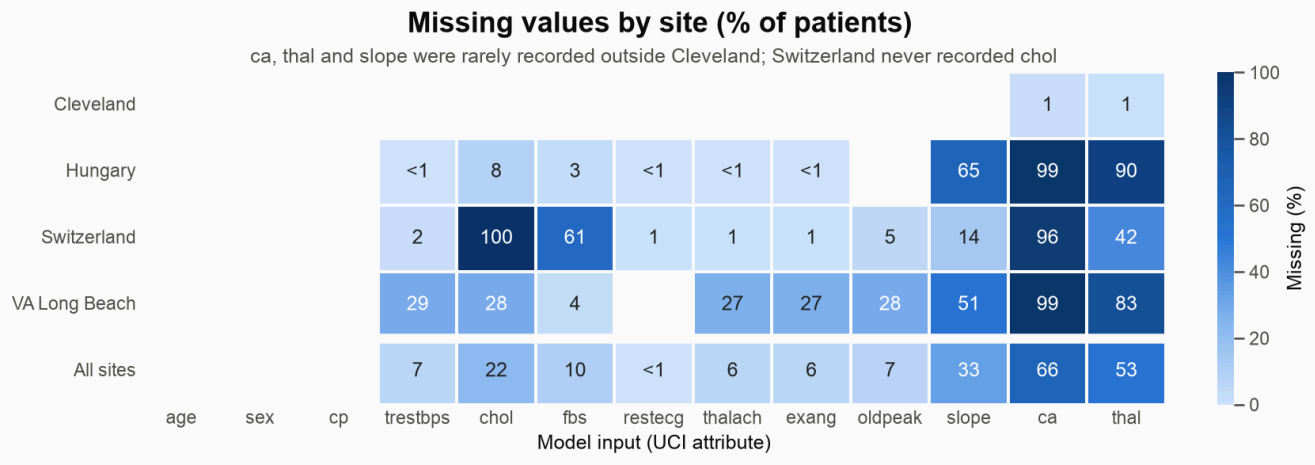


Figure 2. Share of missing values per input and hospital.

- **Missingness is structural (Figure 2).** `ca` is missing for 96-99% of patients outside Cleveland, `thal` for 42-90%, Switzerland never recorded cholesterol, and VA lacks 27-28% of its exercise-test values. Only 299 of 918 rows are complete (297 from Cleveland), so dropping incomplete rows would leave one hospital. Missingness is also informative: the hospital alone ranks patients with ROC-AUC 0.72, and the 35 missingness patterns identify the hospital for 86% of patients. Decision: impute with missing-value indicators, knowing they partly reconstruct the hospital (Section 5.5).
- **Class balance (Figure 11).** 508 patients (55%) have heart disease, so accuracy is meaningful and no resampling is needed, but the rate is 36% in Hungary, 46% in Cleveland, 74% at VA and 93% in Switzerland. Decision: stratify the split and folds on site x class.

- **Distributions and categories (Figures 12, 13).** The exercise-test variables separate the classes best (median maximum heart rate 151 vs 128 bpm, ST depression 0.0 vs 1.1 mm); scales differ by two orders of magnitude. Asymptomatic chest pain (`cp = 4`) has a 79% disease rate against 14% for atypical angina, the codes are not ordinal and "not recorded" has its own rate. Decision: standardise numeric inputs, one-hot encode categorical ones with missing as a category.
- **Correlations and heart rate (Figures 14, 15).** The strongest correlates of disease are exercise angina (r = 0.46), `oldpeak` (0.39) and `thalach` (-0.39); correlation among inputs peaks at 0.39, so no input is dropped by hand. 66% of patients with disease stay below 85% of the age-predicted maximum heart rate, against 40% of those without. Decision: add `hr_achieved_frac`.

5. Feature engineering and model development

5.1 The pipeline

Everything between a raw record and the classifier is one scikit-learn `Pipeline` (`cardiorisk.features.build_pipeline`): (1) `features` keeps the 13 inputs as floats (JSON `null` becomes NaN, extra columns are dropped) and adds `hr_achieved_frac = thalach / (220 − age)`; (2) `preprocess` applies median imputation with missing-value indicators and `StandardScaler` to numeric inputs, most-frequent imputation with indicators to binary ones, and `OneHotEncoder(handle_unknown="ignore")` to categorical ones, with NaN as its own category; (3) `model` is the estimator being tuned. `GridSearchCV` therefore refits imputers and scaler in every fold (no leakage), and the saved object replays the training transforms, with no second copy of the feature code in the API. The indicators also let the API score a record with only `age`, `sex` and `cp`.

5.2 Candidates, validation and selection

Table 3. Model families, their grids and number of candidates n (`train.CANDIDATES`).

Family (fixed settings)	Grid	n
Majority baseline (`DummyClassifier`)	none	1
Logistic regression (`saga`, `max_iter=5000`)	`C` {0.01, 0.03, 0.1, 0.3, 1, 3} x `l1_ratio` {0, 0.5, 1} x `class_weight` {None, balanced}	36
Random forest (`random_state=42`)	`n_estimators` {200, 500} x `max_depth` {4, 8, None} x `min_samples_leaf` {1, 3, 10} x `max_features` {sqrt, 0.5}	36
Hist. gradient boosting (`max_iter=200`)	`learning_rate` {0.03, 0.1} x `max_leaf_nodes` {4, 8, 16} x `l2_regularization` {0, 1}	12

An 80/20 split stratified on site x class gives 734 training and 184 test patients. Five folds of the training split on the same strata are shared by all families, so the comparison is paired. Every fold records ROC-AUC, accuracy, precision, recall and F1 at threshold 0.5. The champion is the non-baseline family with the highest mean CV ROC-AUC, which is threshold-free and insensitive to prevalence; the hold-out is scored once per family, for reporting only. The grids are coarse on purpose: the CV standard deviation is 0.023-0.035 ROC-AUC, while the 18 logistic-regression candidates with `C >= 0.3` span only 0.0029, so a finer grid would select noise.

5.3 Results

Table 4. Model comparison (from `reports/metrics.md`*; CV = mean over 5 folds).*

Metric	Baseline	Logistic regression (champion)	Random forest	Hist. gradient boosting
CV ROC-AUC	0.500 ± 0.000	**0.907 ± 0.029**	0.897 ± 0.023	0.895 ± 0.026
CV accuracy	0.553	0.834	0.823	0.811
CV precision	0.553	0.853	0.829	0.815
CV recall	1.000	0.845	0.857	0.852
CV F1	0.712	0.849	0.843	0.833
Test ROC-AUC	0.500	0.906	0.896	0.900
Test accuracy	0.554	0.842	0.853	0.853
Test precision	0.554	0.829	0.844	0.844
Test recall	1.000	0.902	0.902	0.902
Test F1	0.713	0.864	0.872	0.872

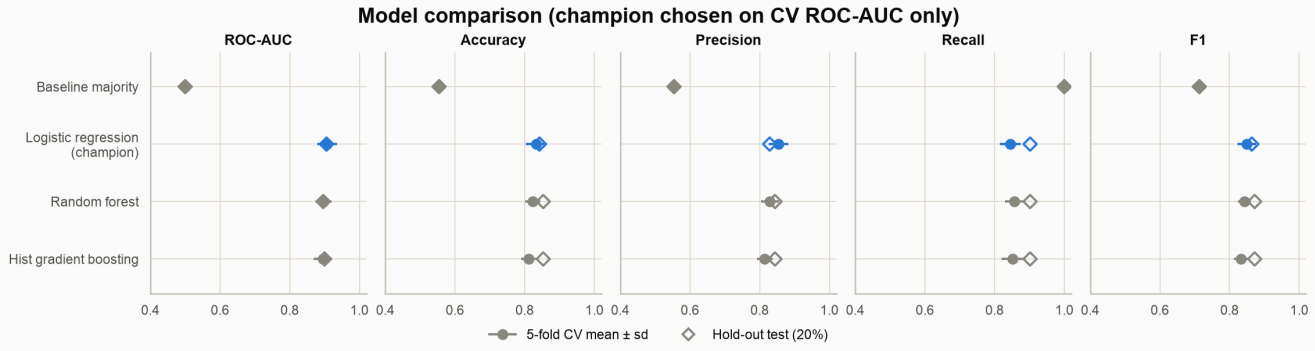


Figure 3. CV mean ± sd (dots) and hold-out score (diamonds) per family for ROC-AUC, accuracy, precision, recall and F1; the champion in blue.

Best settings: logistic regression `C = 0.3`, `l1_ratio = 0.5`, no class weight; forest 500 trees of depth 8; boosting learning rate 0.03 with 4 leaves. All three beat the baseline by about 0.40 ROC-AUC, and CV and hold-out agree within 0.01 (Figure 3), so none is overfitted to the folds.

Why logistic regression. It leads on CV ROC-AUC by 0.010-0.012, less than one standard deviation, so the three are statistically tied and the linear model wins on its merits. Boosting's best configuration used the smallest trees on offer, a sign that extra capacity fits noise at 734 patients; one-hot categories and indicators make the signal close to additive; and the pipeline needs about 205 MiB per pod against 240 MiB for the forest, with coefficients a clinician can read. The ensembles' higher hold-out accuracy (0.853 vs 0.842) is two patients; switching for it would select on the test set.

5.4 The champion on the hold-out set

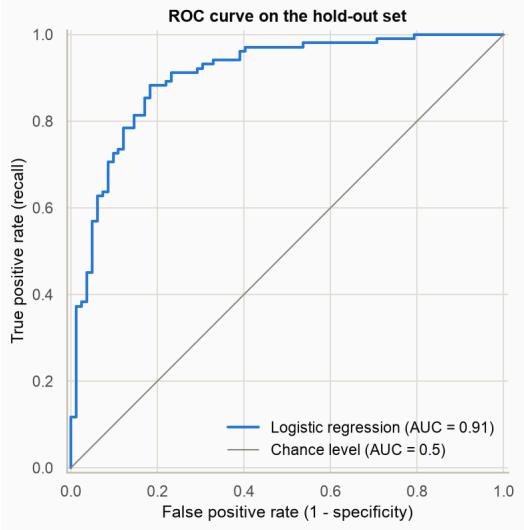


Figure 4. ROC curve on the 184 hold-out patients.

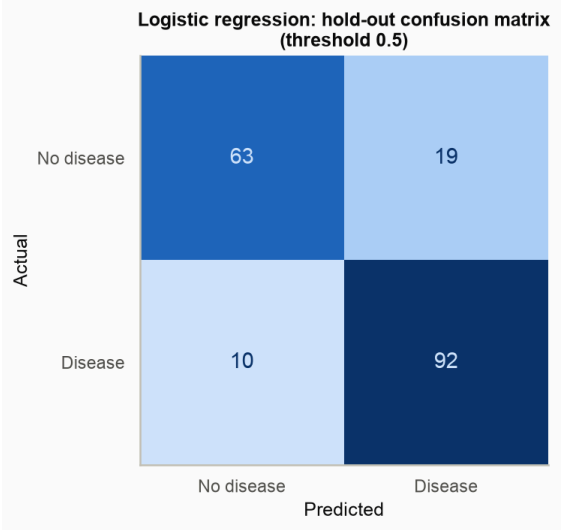


Figure 5. Hold-out confusion matrix at threshold 0.5.

The ROC curve (Figure 4) finds about 80% of disease cases at a 15% false-positive rate. At 0.5 (Figure 5) the model finds 92 of 102 patients with disease (recall 0.902) and clears 63 of 82 without (specificity 0.768): 19 false alarms and 10 missed cases, the right direction for screening. Permutation importance (Figure 16, Appendix B) is led by chest pain type (0.051 ROC-AUC lost), then `ca` (0.018) and `oldpeak` (0.014). `chol` ranks fourth (0.013) despite a correlation of 0.12 because it mostly signals whether cholesterol was measured: shuffling only the recorded values costs about 0.002.

5.5 Leave-one-site-out: how well does the model travel?

Pooled CV mixes all hospitals in every fold. To estimate performance at a new hospital, training refits the champion configuration on three sites and scores it on the fourth (Table 5; chart in Figure 17, Appendix B).

Table 5. Leave-one-site-out results of the champion configuration.

Held-out site	Patients	Disease	ROC-AUC	Accuracy	Recall
Cleveland	303	46%	0.876	0.762	0.878
Hungary	293	36%	0.902	0.795	0.906
Switzerland	123	93%	0.749	0.772	0.791
VA Long Beach	199	74%	0.730	0.759	0.831

Transfer to the two lower-prevalence hospitals is almost as good as the pooled hold-out (0.876 and 0.902 against 0.906), but at Switzerland and VA ROC-AUC falls to about 0.74. Both have high prevalence and the most missing data, a signal a model trained elsewhere reads differently. Accuracy hides the drop: calling every patient diseased would already score 0.74 at VA and 0.93 in Switzerland. The hyper-parameters were chosen by CV over all four sites, so these scores are mildly optimistic; at a new hospital I would expect 0.73-0.90 until the model is validated locally. These metrics are also in the model card, MLflow and `/model-info`.

6. Experiment tracking

MLflow 3.16 tracks every training session in the experiment `cardiorisk-heart-disease`, with a sqlite backend (the registry needs a database) and artifacts under `mlruns/`; CI uploads both as the `mlflow-tracking` artifact. Each session logs four family runs, tagged with `training_session` and `family`, holding the estimator parameters, `cv_<metric>_mean/std` and `test_<metric>` for the five metrics, the training split as an MLflow dataset (734 rows, with digest), plots (ROC, precision-recall, confusion matrix, permutation importance), `cv_results.csv`, a classification report and the model. Under them, 85 nested runs (1 + 36 + 36 + 12), one per grid point, hold the hyper-parameters, CV mean and sd, `fit_seconds` and per-fold

scores (`cv_folds.json`). The champion run adds the tag `champion = true`, the `loso_<site>_*` metrics and the leave-one-site-out plot, and registers a new version of `cardiorisk-classifier` with the alias `champion`.

Nested runs keep the top-level list at four rows while recording every grid point (Figures 18-22, Appendix C). The registry uses an alias instead of the stages MLflow 3 deprecates; moving `@champion` is the promotion step. `model_card.json` and `/health` report the registry version.

7. Model packaging and reproducibility

Serving artifact. `models/cardiorisk_model.joblib` is the fitted pipeline scored on the hold-out; `models/model_card.json` records registry version, hyper-parameters, threshold, metrics, features, MLflow run id, library versions and the SHA-256 of the cleaned table. joblib keeps MLflow out of the image; it can execute code on load, acceptable only because the file comes from the project's own training job and is never accepted from users.

MLflow model. Each family run also logs the model with **skops**, which refuses any type not explicitly trusted; `code_paths` bundles `src/cardiorisk`, so `models:/cardiorisk-classifier@champion` loads without the package.

Requirements and determinism. Both requirement files pin exact versions: `requirements-serve.txt` holds the nine runtime packages the image installs, and `requirements.txt` adds the training and development tools. One seed (42) drives split, folds, estimators and permutation importance, so `make train` reproduces every metric.

8. CI/CD pipeline and automated testing

Tests. 39 pytest tests run offline in about 20 s on synthetic patients, plus one gate on the real data. They cover every cleaning rule and the checksum-verified download, the feature formula and a pickle round trip, the CLI, every API contract (null and omitted fields, 422 without echoed values, `X-Request-ID`, the Prometheus series, JSON logs without patient data) and a reduced `train()` that must write the card, reports, MLflow runs and alias, with logistic regression reaching CV ROC-AUC >= 0.80 on the real data.

Workflow. `.github/workflows/ci.yml` runs on every push to `main`, every pull request and on demand, with six jobs (Figure 6): `lint` (ruff) and `test` (pytest, JUnit XML) in parallel; `train` (data check and full training, model comparison in the run summary, Figure 7); `docker` (builds the image around this run's model, smoke test); `deploy-kind` (a kind cluster with the tested image and the kind overlay, smoke test through the Service, waits until Prometheus reports every API target up); and `publish`.

Gating and publishing. Each job runs only if its predecessors passed, so nothing untested is published; failing jobs print container logs or the cluster's pods and events. On a push to `main`, `publish` tags the image archive that passed both smoke tests `sha-<7>` and `latest` and pushes it without rebuilding, to GHCR with `GITHUB_TOKEN` by default; repository variables and one secret redirect it to Docker Hub, ECR, GAR or ACR.

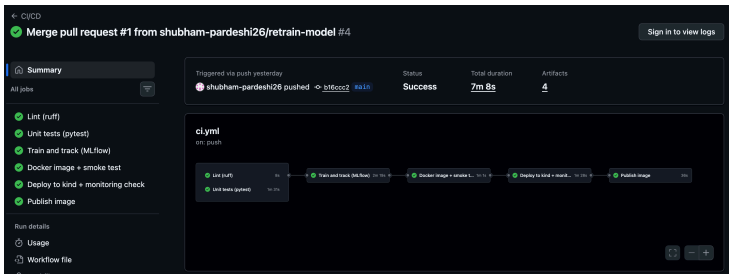


Figure 6. A workflow run with all six jobs green.

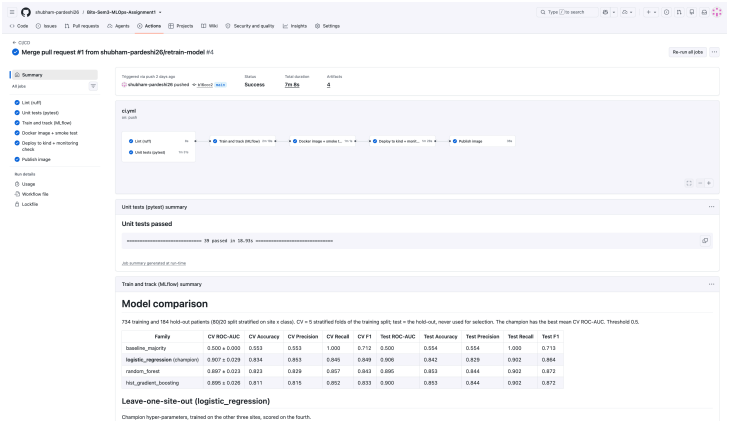


Figure 7. Run summary: unit-test result and the model comparison table.

9. Model containerization

The `Dockerfile` builds on `python:3.12-slim` with only `requirements-serve.txt`, and `.dockerignore` is an allowlist (`requirements-serve.txt`, `src/`, `models/`). Requirements are installed before code and model are copied, so a retrain rebuilds only the last layers. The model is baked in, so one tag identifies code and model. The app runs as uid 10001 with read-only files and a `HEALTHCHECK` on `/health`.

The API (`cardiorisk.api:app`, container port 8000) has five endpoints: `POST /predict`, `GET /health` (liveness and readiness, with the model version), `GET /model-info` (the model card), `GET /metrics` (Prometheus) and `GET /docs` (Swagger UI, prefilled with the sample patient):

```
curl -s -X POST http://127.0.0.1:8081/predict \
  -H 'Content-Type: application/json' -d @scripts/sample_request.json
# {"prediction": 1, "label": "heart_disease", "confidence": 0.9851,
#  "probability_disease": 0.9851, "model_version": "3"}
```

Pydantic rejects codes outside the UCI codebook, implausible measurements and unknown fields, so a misspelt field cannot silently become a missing value; only `age`, `sex` and `cp` are required, and the 422 response does not echo the submitted values. `make docker-build`, `make docker-run` and `make smoke` prove the image (Figures 25-27, Appendix C): the smoke test checks `/health`, the response contract, 422 for invalid payloads and `/metrics`.

10. Production deployment

The manifests use Kustomize. `base/` holds the namespace `cardiorisk`, the Deployment, a ClusterIP Service and the HPA; `monitoring/` adds Prometheus (with a namespaced Role) and Grafana. Three overlays adapt them: `minikube` (image built inside minikube, LoadBalancer on port 8081, nginx Ingress), `kind` (the CI deployment test) and `cloud` (registry image, LoadBalancer and Ingress, notes for GKE, EKS and AKS).

The HPA owns the replica count (2-5 at 70% CPU). Rolling updates keep every ready pod until its replacement is ready (`maxUnavailable: 0`), and all probes use `/health`. Memory is sized from a measured 205 MiB peak (256 MiB request, 512 MiB limit). The namespace enforces the `restricted` Pod Security Standard (non-root, read-only root filesystem, no capabilities).

The deployment in the screenshots runs on minikube with the Docker driver: `make minikube-start` (ingress and metrics-server addons), `make k8s-image` (builds the image inside minikube, so no registry is needed), `make k8s-deploy` (`kubectl --context minikube apply -k deploy/k8s/overlays/minikube`), then `minikube tunnel` in a separate terminal, `make smoke` against the LoadBalancer and `make smoke URL=http://127.0.0.1` against the Ingress (Figures 8, 9). For a managed cluster only the lines in `overlays/cloud` (image, Ingress class) change.

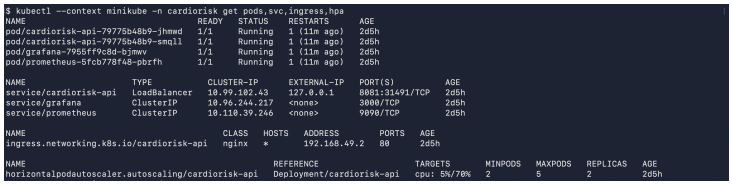


Figure 8. `kubectl --context minikube -n cardiorisk get pods,svc,ingress,hpa`.

```
$ make smoke
.venv/bin/python scripts/smoke_test.py http://127.0.0.1:8081
Smoke testing http://127.0.0.1:8081
PASS  health: model_version 1
PASS  predict sample patient: heart_disease, p=0.9851
PASS  X-Request-ID echoed
PASS  422 for missing age: HTTP 422
PASS  422 for unknown chest pain code: HTTP 422
PASS  422 for cholesterol out of range: HTTP 422
PASS  422 for unknown field: HTTP 422
PASS  metrics expose cardiorisk_predictions_total
SMOKE TEST PASSED
$ make smoke URL=http://127.0.0.1:8088
.venv/bin/python scripts/smoke_test.py http://127.0.0.1:8088
Smoke testing http://127.0.0.1:8088
PASS  health: model_version 1
PASS  predict sample patient: heart_disease, p=0.9851
PASS  X-Request-ID echoed
PASS  422 for missing age: HTTP 422
PASS  422 for unknown chest pain code: HTTP 422
PASS  422 for cholesterol out of range: HTTP 422
PASS  422 for unknown field: HTTP 422
PASS  metrics expose cardiorisk_predictions_total
SMOKE TEST PASSED
```

Figure 9. minikube tunnel and make smoke passing against the LoadBalancer and the Ingress.

11. Monitoring and logging

Metrics. The API exports `cardiorisk_http_requests_total` (by method, route and status) and `cardiorisk_http_request_duration_seconds` for traffic, errors and latency; `cardiorisk_predictions_total` by label and `cardiorisk_prediction_probability` (a coarse drift signal) for the predictions; and `cardiorisk_model_info` with the version each pod serves. The `route` label is the route template, so random URLs cannot create unbounded series. Prometheus discovers annotated pods and scrapes every 5 s (Figure 28, Appendix C). Grafana is provisioned from ConfigMaps; the "CardioRisk API" dashboard (Figure 10) shows pods up, model version, prediction mix, request rate, error share, `/predict` latency p50/p95/p99, memory and the predicted-probability distribution. `make load` sends 300 varied requests, 10% invalid, so every panel has data.

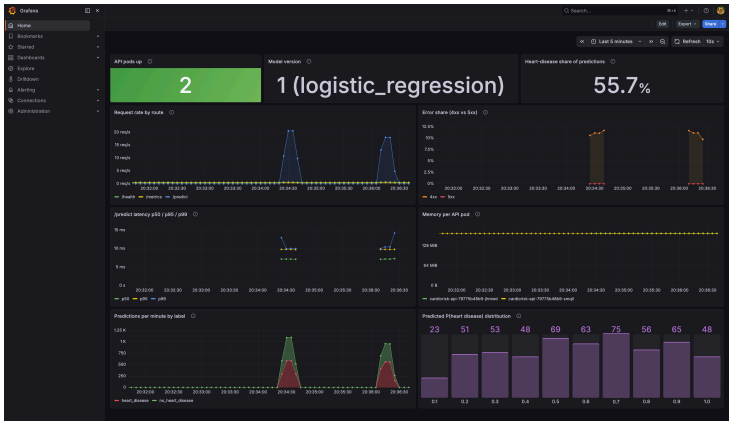


Figure 10. The Grafana "CardioRisk API" dashboard with traffic from make load.

Logs. Every request writes one JSON line to stdout (`event`, `request_id`, `method`, `route`, `status`, `duration_ms`) and every prediction one more (`label`, `probability_disease`, `model_version`), joined by the request id: the caller's `X-Request-ID` or a generated one (Figure 29, Appendix C). Probes and scrapes are logged at DEBUG, so the INFO log shows real traffic. Patient features are never logged, which a test enforces.

12. Limitations and future work

- **Hospital shift.** Leave-one-site-out ROC-AUC drops to 0.73-0.75 at the two high-prevalence sites, and part of the pooled score comes from recognising a hospital by its missing tests. Next: train without missing-value indicators and compare, and tune by nested leave-one-site-out CV.
- **Calibration and threshold.** Probabilities fitted to the pooled 55% mix are likely off at 93% and 36% sites; training should report the Brier score, sites should recalibrate, and the 0.5 threshold should come from a recall target agreed with clinicians.
- **Drift.** The probability histogram is a coarse proxy; per-input distributions and missing rates should be compared with the training data, and confirmed diagnoses collected.
- **Data and tracking.** Checksum pinning suits a static dataset; a growing one needs DVC or remote MLflow storage and a shared tracking server.
- **Security and operations.** No authentication, TLS or rate limiting (coursework, not a medical device); Prometheus data lives on an `emptyDir`, and the HPA scales on CPU only.

Appendix A. Repository structure

```
.
├── src/cardiorisk/          config, data, eda, features, train, predict, api
├── test/                    pytest suite (39 tests) and conftest.py
├── notebooks/01_eda.ipynb   EDA with commentary
├── scripts/                 smoke test, sample request, architecture diagram
├── data/raw/                four UCI site files + codebook (SHA-256 pinned)
├── data/processed/          heart_disease_clean.csv (918 rows)
├── models/                  cardiorisk_model.joblib, model_card.json
├── reports/                 REPORT.md, report.docx/.pdf, metrics.*, figures/
├── screenshots/             evidence for this report (01–15)
├── deploy/k8s/              base/, monitoring/, overlays/{minikube,kind,cloud}/
├── deploy/compose/          prometheus.yml for docker-compose
├── .github/workflows/       ci.yml
├── Dockerfile, .dockerignore, docker-compose.yml, Makefile
└── pyproject.toml, requirements.txt, requirements-serve.txt
```

The command reference (every `make` target and variable) is in `README.md` and `make help`.

Appendix B. Additional figures

The EDA figures of Section 4.2 (larger and with commentary in `notebooks/01_eda.ipynb`), the champion's permutation importance (Section 5.4) and the leave-one-site-out chart (Section 5.5).

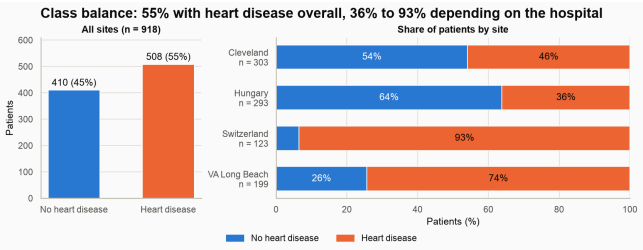


Figure 11. Class balance overall and by hospital.

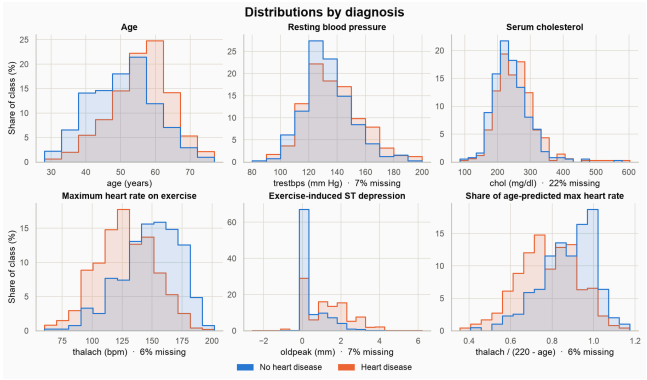


Figure 12. Numeric inputs by diagnosis.

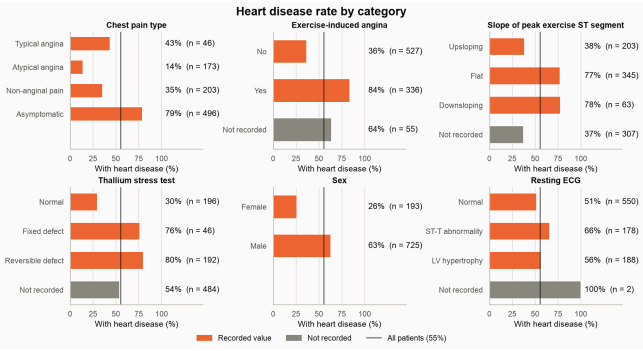


Figure 13. Disease rate per category, "not recorded" as its own bar.

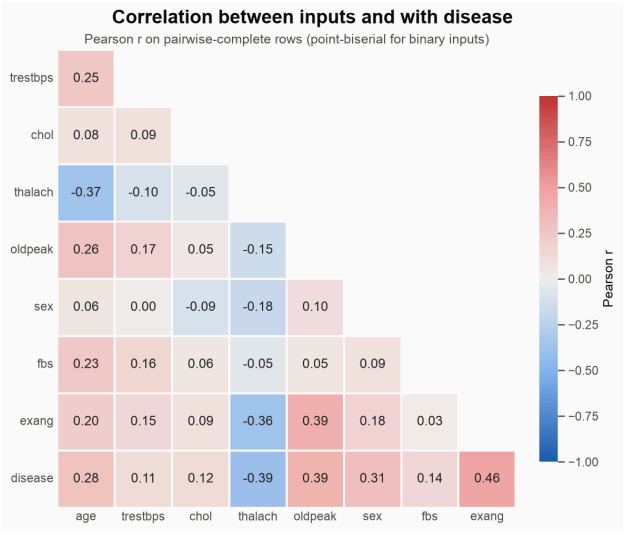


Figure 14. Pearson correlations between the inputs and with disease.

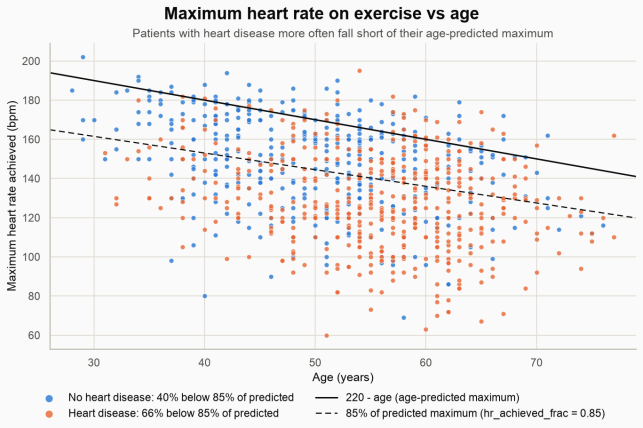


Figure 15. Maximum heart rate against age, with the line 220 - age.

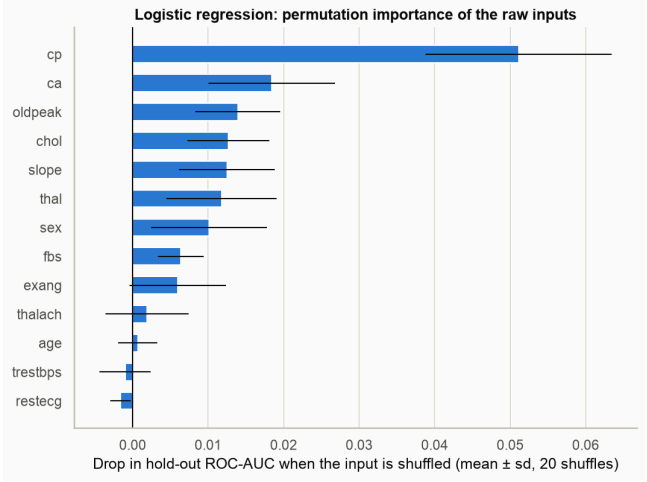


Figure 16. Permutation importance on the hold-out (20 shuffles).

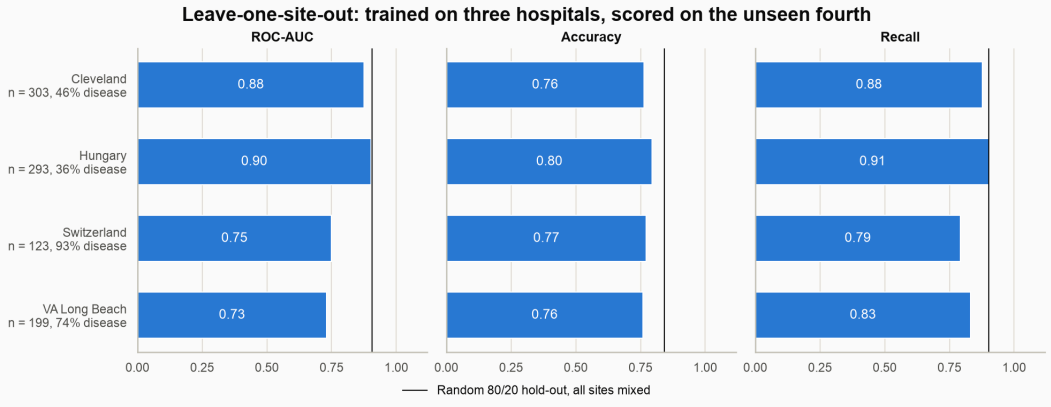


Figure 17. Leave-one-site-out scores of the champion configuration; the vertical line is the random 80/20 hold-out.

Appendix C. Additional screenshots

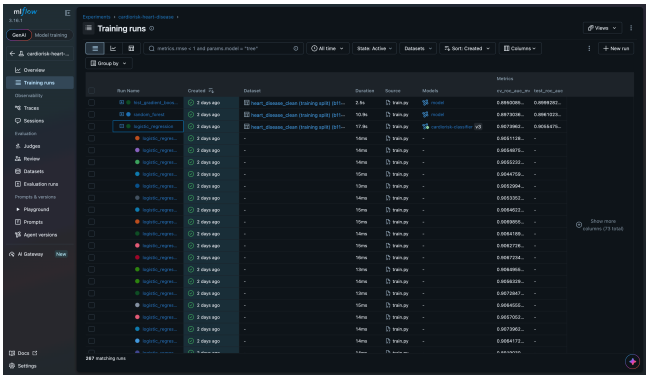


Figure 18. MLflow experiment: the four family runs of a session.

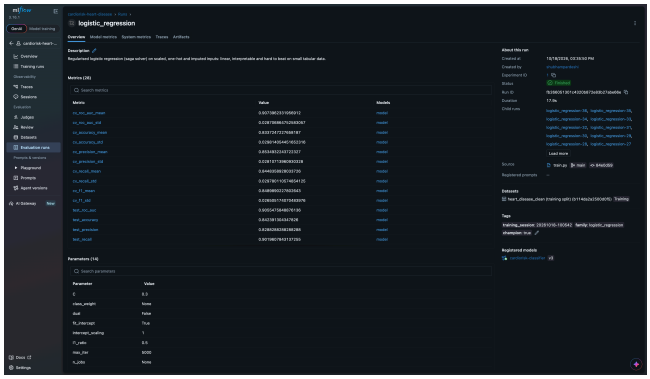


Figure 19. MLflow champion run: metrics and parameters.

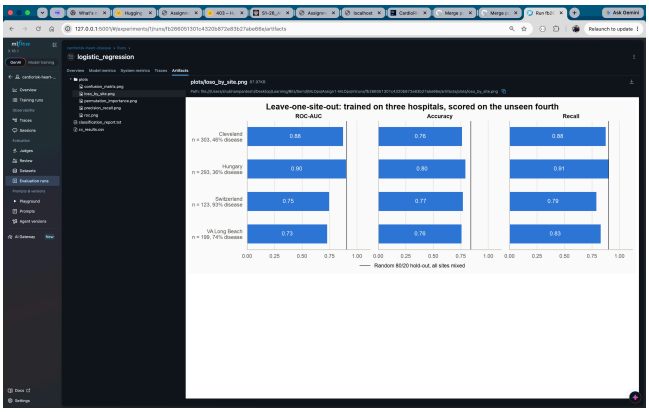


Figure 20. Champion run artifacts: `plots/`, `cv_results.csv`, `report.`

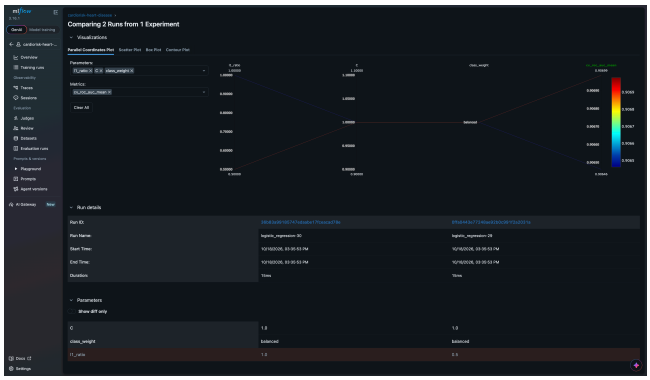


Figure 21. Compare view of the 36 nested logistic-regression runs.

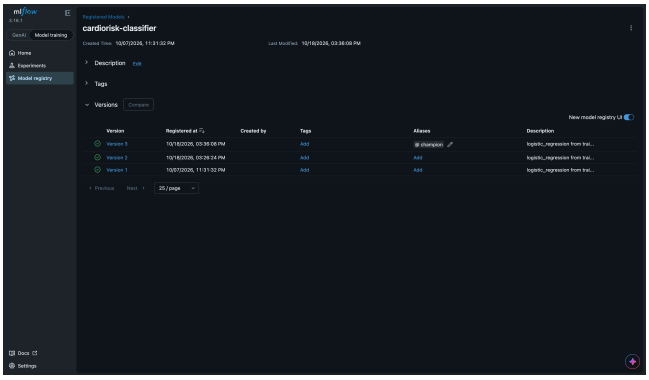


Figure 22. Registry: `cardiorisk-classifier@champion`.



Figure 23. Artifacts of the workflow run.

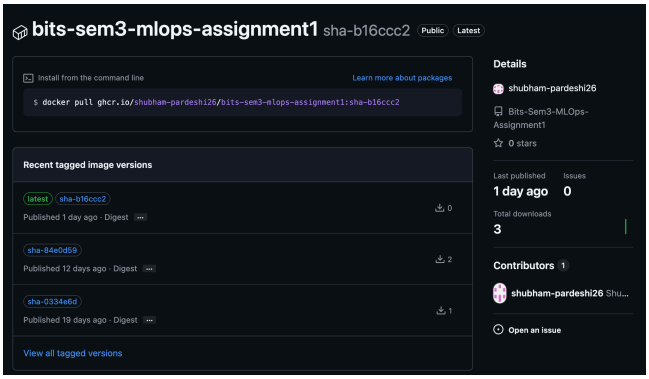


Figure 24. GHCR package, tags `sha-<7>` and `latest`.



Figure 25. `make docker-build` and `make smoke` passing.



Figure 26. `make docker-run`*: the container's JSON logs.*

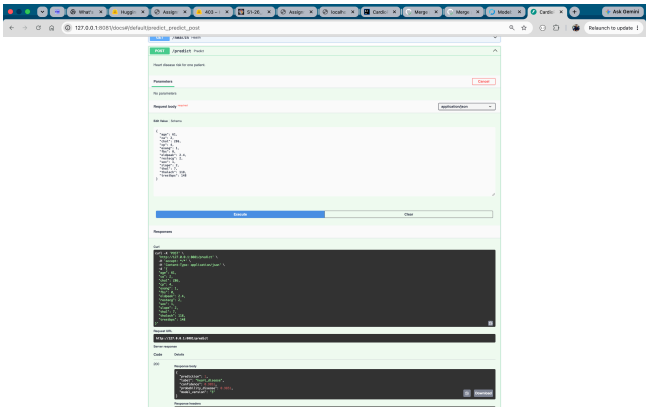


Figure 27. Swagger UI: a `/predict` *request and response.*

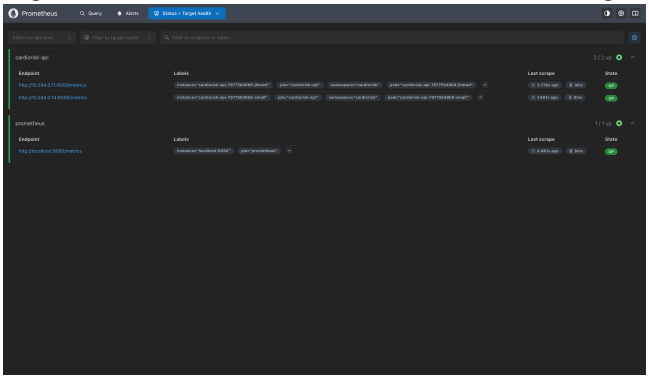


Figure 28. Prometheus: `cardiorisk-api` *pod targets UP.*



Figure 29. `kubectl logs`*: JSON request and prediction lines.*